Rady | School of Management
UNIVERSITY OF CALIFORNIA SAN DIEGO

MGT 452R: NEW PRODUCT DEVELOPMENT

Executive Weekend MBA
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COURSE DESCRIPTION

This course is a foundational elective to continue shaping your MBA for a career based upon Innovation, Design, or Entrepreneurship. Along with the prerequisite course *MGT 412/MGT 412R: New Venture Design*, it forms the core of the Certificate in Design & Innovation. For those looking to fully prepare their startup to launch post-MBA, it provides a deep exploration of the product / service they will bring to market, and is complimentary to *MGT 415: New Venture Development,* which provides a deep exploration of the business' marketing, finance, and operations.

The New Product Development (NPD) course is based upon the instructor's experience as a practitioner, and with the view that the student's role after graduation will likely be that of manager / participant / facilitator of NPD efforts. To that end, we will work to give the student a broad framework of the process, as well as specific tools and techniques that can be used along the way.

We will take a user-centered approach to NPD, drawing heavily on the idea of empathic design to gain a visceral understanding of what people want in a product. The emphasis will be on qualitative (vice quantitative) research techniques aimed at uncovering the "big picture" of what new product you should be developing, as well as for detailing form, features, and functionality of a specific product under development.

Our goal is to be able to define the ecosystem in which product and user live, so we can craft a complete solution that is more valuable to consumers and more difficult for competitors to dislodge, much like Apple's iPod / iTunes versus standalone MP3 players. This approach should also make our new product vision more explicit and more easily communicated to others in the organization, to gain resources and rally support.

The course follows a **"hybrid" online + in-person format**, with three online modules in Canvas, and three in-person class sessions at Rady, alternating online / in-person. Additionally, teams will meet individually with the instructor between the second and third in-person sessions, for focused coaching and critique as they design and develop their final project products and final project presentations.

Course materials are a mix of case studies, non-business-text readings (such as the works of Edward Tufte), multimedia (such as the films of Ray and Charles Eames), and practical exercises. Active participation is vital, as much of this can only be learned by doing.

There is a final project, in which student teams will craft a product and the accompanying explanation of why this is a compelling product for both user and business. We will use information from the practical exercises as partial input to the process, though outside supplemental work is expected. All teams will have broad latitude to choose the actual product they develop as their final project, though it must incorporate both a physical and digital component, in order to solidify their understanding of and ability to engage in current NPD practice.

COURSE OBJECTIVES

1. Understand the overall timeline / framework for NPD.
2. Understand and be able to apply user research techniques to inform the development process. Be able to abstract the data into useful insights, to uncover feature requirements and define the ecosystem in which the product will live.
3. Understand and be able to apply generative techniques to define and refine the core product and extended product.
4. Understand and be able to apply prototyping and visualization techniques to communicate the product vision.
5. Understand organizational issues in NPD

COURSE MATERIALS

The course will draw upon **Ulrich and Eppinger's textbook** to provide an understanding of the basics of new product development, and upon Don Norman's foundational book **The Design of Everyday Things** for a solid grounding in Cognitive Design. Cases, articles, and extracts are provided in the course reader as the basis of class discussion. We will also draw from other **materials listed under the "Required" list below**. Additional materials will be distributed as required for practical learning exercises.

Required

- Product Design and Development, 7^{h} Edition, Karl Ulrich & Steven Eppinger, McGraw Hill, 2011
- The Design of Everyday Things, Revised and Expanded Edition, Don Norman, Basic Books, 2013
- The Market Research Toolbox: A Concise Guide for Beginners, 4^{th} Edition, Edward F. McQuarrie, Sage Publications
- Textbooklet: Visual and Statistical Thinking: Displays of Evidence for Making Decisions, Edward Tufte, Graphics Press
- Essay: The Cognitive Style of Powerpoint, Edward Tufte, Graphics Press

  Note: The Market Research Toolbox was substantially rewritten in the fourth edition. If you try to use an earlier edition, you will miss material, and the chapter assignments will not match up.

Course Reader

- Henry Heinz: Making Markets for Processed Foods, HBS 9-801-289
- "Spark Innovation Empathic Design," Dorothy Leonard and Jeffrey Rayport, Harvard Business Review, Nov-Dec 1997, #97606
- "Naked Truth Meets Market Research: Perfecting a New Shower Head? Try Watching People Shower." Washington Post Article, Sunday, February 24, 2002. http://www.washingtonpost.com/wp-dyn/articles/A55864-2002Feb23.html
- "Margaret Mead Meets Consumer Fieldwork," Jennifer McFarland, Harvard Management Update, August 2001, Product U0108C
- Sony AIBO: The World's First Entertainment Robot, HBS 9-502-010
- Innovation at 3M Corp. (A), HBS 9-699-012
- Note on Lead User Research, HBS 9-699-014
- "Customers as Innovators: A New Way to Create Value," Stefan Tomke & Eric von Hippel, Harvard Business Review, April 2002, Product R0204F
- "Enlightened Experimentation: The New Imperative for Innovation," Stefan Tomke, Harvard Business Review, Feb 2001, Product R0102D
- "R&D Comes to Services: Bank of America's Pathbreaking Experiments," Stefan Tomke, Harvard Business Review, April 2003, Product R0304E

Resources

If you are thinking of building a basic bookshelf to bolster your NPD skills, here is my recommendation. It covers a wide range of topics, reflecting the cross-functional nature of NPD, and focuses on the aspects that will give you a competitive advantage, as opposed to the aspects that are well-understood. The links to amazon.com are for information only.

Designing for the Digital Age: How to Create Human-Centered Products and Services
Kim Goodwin
A practical guide to leading, managing, and doing design, from the perspective of a principal at one of the world's foremost design firms. Applicable to both agency and internal design groups.

The Evolution of Useful Things
Henry Petroski
How some common items got to be the way they are - the changes they went through, and why these changes happened.

Middletown: A Study in American Culture
Robert and Helen Lynd
A rich study of the state and dynamics of turn-of-the-century America - a reference piece. Instructive in its content, as well as its blending of objective, subjective, quantitative, and qualitative information to give a complete picture.

Guns, Germs, and Steel
Jared Diamond
A reference example of thinking and explaining in an insightful and analytical manner. Takes a complex issue of human societal development, and goes beyond the usual superficial analysis to find root cause and drivers.

Ethnography
Martyn Hammersly and Paul Atkinson

A primer on what it is and how to do it.

Customer Visits
Edward McQuarrie
In the same vein as Toolbox (written for non-marketing folks - R&D managers, engineering or ops managers, etc.), on a single tool. Much of what he says about how to speak with customers to elicit useful, less-biased information is broadly applicable to other methods of inquiry.

How to Design Surveys
How to Ask Survey Questions
Arlene Fink
Quick reads, aimed at people creating quantitative studies, but the principles of research design and question construction apply to qualitative inquiry as well.

Managing the Professional Service Firm
David Maister
If you are running an R&D, design, or engineering group within a company, you would do well to think of yourselves as a professional services firm. If you are buying services from an outside firm, a good guide to what they are (or should be) doing, and where their interests lay.

The Innovator's Dilemma
The Innovator's Solution
Clay Christensen
Discusses the concept of "good enough" with respect to users' needs and desires, and the dynamics of competition this imposes on products.

Wellsprings of Knowledge
Dorothy Leonard
Discusses the concept of empathic design, and how innovative organizations build that capability.

The Visual Display of Quantitative Information
Envisioning Information
Visual Explanations: Images and Quantities, Evidence and Narrative
Edward Tufte
The definitive works on using visuals to promote clear thought and clear communication.

Flatland: A Romance of Many Dimensions
Edwin A. Abbott
A fun read, and a good exercise in thinking about how a society's ability to perceive shapes its activities, its views on what is "normal," and its capacity for understanding other cultures.

Course Assignments

See grading criteria and schedule.

COURSE GRADING

Your performance will be evaluated as follows:

Component	Percentage
In-Person and Online Participation	30 %
Practical Exercises and Submissions	30 %
Team Project Presentation (In-person at the final class session)	40 %

Class Participation: Your class participation (contribution to class discussions) will count for 30% of the class grade. To help in preparation, I have suggested "Guiding Questions" for every class in the detailed syllabus. These questions should serve merely as a starting point. At the beginning of the class, one or more class members will be asked to start the session by addressing a specific question, or by providing an overview of the reading / preparatory exercise. Anyone who has prepared the case or done the exercise should be able to handle such a leadoff assignment. After a few minutes of initial analysis and recommendations, we will open the discussion to the rest of the class. As a group we will try to build a complete analysis of the situation. You are expected to be an active participant throughout the entire class and to contribute to the quality of the discussion. Please note that the frequency (i.e., the quantity) of your interventions in class and in online forums is not a key criterion for effective class contribution. Instead, we focus on quality. Some criteria used to evaluate class contribution are as follows:

- Is the participant an active listener, looking for opportunities to enrich the discussion?
- Are the points that are made relevant to the discussion? Are they linked to the comments of others?
- Do the comments show evidence of analysis of the case / reading / exercise?
- Is there willingness to test new ideas, or are all comments "safe"? (For example, repetition of case facts without analysis and conclusions.)
- Do comments clarify or build upon the important aspects of earlier comments and lead to a clearer understanding of the case / reading / exercise?

Practical Exercises: New product development is cross-functional, time-sensitive, and competitive. As a result, leading-edge techniques are constantly being generated and refined, and most practitioners learn by doing. We will use individual and team exercises as a teaching tool, and to provide information for the final team project. There is no “right” answer for these, but because some of the exercises require outside prep, and because the individual contributions form the basis of a collective learning experience, I will be evaluating your preparation, intensity of application, and thoughtfulness of execution.

Attendance: In a hands-on, project-based class, it is very difficult to make up for an absence, because so much of the learning comes from your engagement with the activity in the moment, and your reflection upon it afterwards. You are expected to attend every class, and you are responsible for the material covered in class whether or not you attend. However, I realize that despite your best efforts you might be forced to miss a class due to unforeseen emergencies. Makeup for the session will depend upon the material covered. Because of our reliance on practical exercises and team work, you should realize that an absence affects classmates as well. Given that we have only three in-person sessions, missing any one will seriously degrade your learning experience.

Team Project: There will be a team deliverable due the last in-person class session which will count for a total of 40% of the class grade. It is intended to take you through a new product development cycle, though it is compacted to be manageable within the constraints of a university course and your schedules. The general topic will be the same for all teams, and the practical exercises will serve to provide some baseline information to inform the development of this new product.

I will expect your team to:

- further supplement this with additional research,
- synthesize an understanding of the environment of use,
- uncover opportunities and imperatives for a new product / service in that space,
- prototype critical elements of the new product / service, both physical and digital, and
- communicate to the rest of us the vision for the product, and why it is compelling for both user and business.

The deliverables are in two parts, graded separately:

- A 10 to 20 minute in-class presentation to the class, and
- A report on the project in whatever medium the team desires.
- Please remember that quality, not quantity, is the key to success in the project.

We may set aside some class meeting time for team project work, and I will be available for consultations and work sessions with the teams. I am happy to help guide and critique your work along the way.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
https://academicintegrity.ucsd.edu/process/policy.html

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the OSD website for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.

ARTIFICIAL INTELLIGENCE (AI) POLICY AND APPROACH

Generative AI is a disruptive force in both business and education. While many choose to focus on the negative aspects of disruption, leaders of Design and Innovation teams recognize that the disruption of established systems and competitive dynamics also creates opportunities for those who are able to harness the new tools and master the new landscapes created as a result. This is the essence of the concept "Disruptive Innovation" at the heart of Clayton Christensen's seminal work *The Innovator's Dilemma*.

We also know from the history of technological advances in design and manufacturing that new tools tend to improve both quality and speed of work for the thoughtful user, and shift the thoughtful adopter's focus to the higher-value creative and intellectual tasks. Thoughtless (and non-) adopters become obsolete.

The assignments in this course do not lend themselves to simply "having ChatGPT do it;" I know this from experiments we have run in previous quarters of this course. There are, however, elements that we can learn to accomplish better or faster by treating generative AI as an assistant to our work, which further helps us develop our ability to critique and guide innovation work, critical managerial skills. There are also examples of good process that a custom agent can model for us to help us learn to use good process. You will have an opportunity to experience this firsthand through exercises during our in-person sessions and through custom agents I have created for your use in this course, such as the *NPD Ideation Assistant*.

To help us advance our ability to use AI as an effective tool and to identify ineffective uses of AI, I allow its use in this course under the following conditions: **Cite and Critique**.

Cite: Identify which AI tool you used, and for what task(s). Save and forward the prompt and output.

Critique: Give your assessment of how helpful the tool was to you and your team. Identify how it could have been more helpful. I will synthesize these learnings for the class as a whole.

This applies both to any tools you discover and to the custom tools and agents I supply for your use.

I will supply at least two custom agents for use in this course, specifically designed to assist you in learning the fundamentals of NPD and pushing your own practice to be more effective. They are best thought of (and interacted with) as teammates:

> <u>Ideation Assistant</u>:
> You initially encountered this agent in MGT 412, as an aid to brainstorm product concepts based upon an understanding of a particular user, need, or context. It is particularly effective in pushing teams to explore a larger volume of ideas than MBA student teams usually pursue.
>
> <u>Visualization Assistant</u>:
> Most people believe they cannot draw. This is wrong. We can all draw sufficiently well to illustrate our thinking. Further, visual communication is an essential act of leadership for managers working with design and engineering teams; your people communicate this way, and they will expect you to be comfortable doing so as well.
>
> The visualization assistant is meant to provide a final level of detail to your own sketching efforts. You may use it in (almost) any assignment for this class, and when you do so, you must also provide

your own initial sketches that you fed into the assistant. You **may not** use this or any other AI tool in the Design-a-Food exercise.

COURSE SCHEDULE

The course is arranged to generally follow the flow of activities in a typical new product development cycle, with some adjustment to allow for exercises that must be prepared ahead outside of class. It is "front-loaded," to ensure that students have the tools needed for the final project. This gives us some flexibility in the final sessions to explore specific topics in greater detail. Depending upon feedback from the class, we can explore topics such as organizational issues for large / small / entrepreneurial NPD, process differences in software and biotech/pharma, etc. I actively solicit your input on this.

Videos, readings, and online exercises are posted on Canvas, arranged in modules that you must complete before the in-person session that concludes each module. Allow some time for this – it is impossible to "binge watch" the day before and successfully complete the online exercises.

Setup	Topics and Preparation
Team Formation	Most of the assignments in the first Online Module are individual assignments. The **one exception** is the *Food Observation and In-Context Interview*, which you conduct as a research-team-of-two with another student, and the *Design-a-Food Exercise*, in which your research-team-of-two works with another research-team-of-two as a design-team-of-four, emulating how NPD teams work in actual practice. I have set up an online forum to help you match up for both of these exercises. You will start working in your final project teams during Online Module 2. However, you should not delay forming that team, as there is much to accomplish together between the first and second in-person sessions. Remember that the team you formed in MGT 412: *New Venture Design* was only for that quarter. If you wish to continue developing your concept from that course into an actual product in this course, you may choose to continue working together, or you may choose to form or join another team. Or, you may choose to work on something completely different. That is entirely up to you. I have also set up an online forum to help you match up in a final project team.

Online Module 1	Topics and Assignments
Course Intro	• View my video welcome to the course. • View the three videos on Canvas exploring the development and impact of the Jobs-to-be-Done approach to product development.

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Read chapters 1-3 in Ulrich and Eppinger’s canonical text Product Design and Development

If this seems like a lot of video to watch, please know that we follow best practices of online coursework, limiting videos to a maximum of 5-10 minutes, with activities or readings between the videos to reinforce the learnings.
As you watch and read, consider the following framing questions:

Have you ever been part of a product development effort? What was the team structure? What was the project structure? What did you think worked well? What do you think worked poorly?
What do you think of the project structure Ulrich and Eppinger lay out in Ch 2? Does it make sense? Do you see anything missing?
Is there a “most important” step in the development process?
Why or why not?
What do you think of the product planning process described in Ch 3? Would you characterize it as thorough or random?
Customer-led or company-led?
What do you think of the “Jobs to Do” approach described in the videos?
Define for yourself the term “product.”
What exactly does it encompass?

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<td>Film:
A Computer Glossary</td>
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View my video setting the context of the A Computer Glossary film.
View the film A Computer Glossary. As you watch, consider these questions from the perspective of a mid-level manager in the late 1950’s:

What is my first reaction to this film?
Are there elements in the film that would help me answer:

Does this new device present opportunities for us to expand / scale our business operations?
What sort of investment would be required?
How does this all work?

Contribute your reactions, thoughts, and questions to the online discussion forum. Read through your classmates’ contributions and engage in discussion.
As you engage in discussion, consider and contribute:

Is there an overarching theme(s) in the discussion?
What have the designers accomplished with this film?
Can you pull out specific examples from the film, of techniques that seemed particularly effective?
It may help to consider:

What they have communicated in this part of the film?
How have they communicated this?
Why was it important to communicate this point in this way?

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	• View my video wrap-up, summarizing the useful elements in the film and how they might provide what is missing in the New Product Development process as practiced by most organizations.
"Meh" - Just an Average Product	• Choose a product from your everyday life that you use regularly, but neither like nor dislike. o Using the actual product as a visual aid, record a video for a maximum of two minutes on exactly what job it performs for you and why your overall impression of it is "meh." o Consider why you keep this product around and keep using it. o Why have you not replaced it? o What would you do if it broke? • Post your video to the online discussion forum, and view at least five of your classmates' videos. • Contribute your thoughts and observations to the online discussion forum. Read through your classmates' contributions and engage in discussion. • View my video wrap-up of the exercise, once you have had a few days for online discussion with your classmates..
Psychology of New Product Adoption	• View my video introducing and explaining the issues surrounding customers' perceptions of the costs and benefits of purchasing a new product, or replacing an existing product with a (supposedly better) competitor. • Pay particular attention to the typical mismatch between the perspective of the customer and the perspective of the development team. • Consider what that means for the manager leading a development project.
In-context Observation and Interview: Food	• Pair up in teams of two to plan and conduct an Observation and In-Context Interview with a third non-Rady person as the participant (I suggest not your significant other), capturing their relationship with food – acquisition, preparation, consumption, recovery. Be sure to capture both functional and emotional elements. • Begin by constructing an interview protocol for your team – mind-mapping is a good tool to help you surface your own understanding, suppositions, and blind spots – so that you can know what activity you want to observe (preparation and consumption), what you expect to see, what might be surprising, and what you just don't know anything about. • This builds upon the food photo exercise in MGT 412, extending your skills to become a true field researcher, which is foundational for anyone seeking to practice User-Centered Design. • Capture your findings however you think best. You will pair up with another team to share and further synthesize your findings, and use them as the foundation of another exercise.
Kano Model	• View my video explaining the origin, content, and application of Kano's model of customer satisfaction. • Pay particular attention to the definitions of the three categories of features, and how they combine to forecast a customer's "happiness" with a product.

	• Consider the implications for a manager seeking to match their team's development efforts with their organization's strategic goals and competitive positioning. • What do you think of the "Minimum Lovable Product" approach?
The "Insights → Qualities → Features → Sketch" Framework	• View my video introducing and explaining this structured approach to deriving product features from the user research that a development team conducts. • Pay particular attention to the utility this provides a development team when it comes time to present, explain, and defend their detailed design decisions for the product. • Consider what that means for the manager working with stakeholders outside of the core development team. • You will use this framework as the foundation of the Design-a-Food-Product exercise.

Design-a-Food-Product Exercise	• Working as a team of 4 (your two pairs that shared your food research with each other), continue the exercise we began in class last week.
	• Design a food product for one of the users that you studied. While we normally design for a composite persona that represents a segment of users, in this case we are designing for the specific user you studied, to practice designing from data.
	• Use the Insights → Qualities → Features → Sketch framework that we learned and explored in the video on Canvas, taking care to describe and show the relationship between the qualities that this user would find desirable in the product, and the features that deliver on that quality.
	• There are two deliverables for this assignment:
	  o A brief summary of your insights and understanding of the user for whom you are designing this product.
	    ▪ for this, you do not need to upload anything - you may simply talk through your existing user research materials in class
	  o A sketch and explanation of the product you have designed.
	    ▪ for this you do need to upload your sketch and explanation
	  o You may use a maximum of 2 slides for your presentation, one of which must be a hand-drawn sketch of the product.
	  o Each team will have 5 minutes to present and receive critique.
	  o The use of AI to produce the sketch is specifically prohibited. It must be hand-drawn by the team.
	• Upload your product on Canvas, and be ready to present and engage in critique with classmates in our in-person session.

Heinz Case	• Thoroughly prepare, and be ready to discuss when we meet for our first in-person session, the case: *Henry Heinz: Making Markets for Processed Foods*, HBS 9-801-289 • As you prepare, consider the following questions: o If you were Henry Heinz in 1865, with half interest in a brickyard, and three or four acres of farmland producing horseradish and

	miscellaneous vegetables, how would you think about opportunities for the future? o If you knew you wanted to pursue your food business, how would you figure out what to make? o What is it that Heinz is really selling? o How did Heinz manage to create this thing that he is selling? o How well or poorly does it fit into his customers' lives? o Is this a story of brand creation or product development? • Also read the following short articles, and consider how they apply (or not) to the Heinz case, and to the *Observation and In-Context Interview* exercise: o "Spark Innovation Through Empathic Design," Dorothy Leonard & Jeffrey Rayport, *Harvard Business Review*, Nov-Dec 1997 o "Naked Truth Meets Market Research: Perfecting a New Shower Head? Try Watching People Shower." *Washington Post*, Sunday, February 24, 2002
Team Formation	• Participate in the online forum as an aid to finding teammates and forming a final project team to work on team exercises through the quarter, leading to the final project and final presentation. The ideal team size is 4-5 people.

In-Person Session 1 End of Week 1/2	**Topics and Preparation**
	We will provide details and guidance, and lead exercises and discussions live in the classroom. Be sure that you have completed all of Online Module 1.
Team Formation	By now you should have a team to work with on your final project over the course of this quarter. This does not have to be a continuation of your work in *MGT 412: New Venture Design*, though it may be if you so choose. You may change topics and/or teams when you begin the Winter quarter.
What is New Product Development?	We will discuss both your thoughts and your interests in the topic, so that we may customize our approach to New Product Development to best support your career goals. Be sure that you have viewed / reviewed the videos and readings in Module 1.
Design-a-Food-Product Presentations and Critique	We have already learned the value of a structured approach to critique and iteration. You will present the first iteration of your food product, and critique the work of your classmates. We will then begin an iteration cycle to improve upon your work using the concepts and frameworks of cognitive design.

Milk Carton Exercise	We will conduct an in-class exercise to explore the practical applications of cognitive design and its use in the New Product Development process.
Heinz Case Discussion	We will discuss the case *Henry Heinz: Making Markets for Processed Foods*, HBS 9-801-289, and use it to explore one of the two fundamental approaches to New Product Development. Be sure that you have prepared for the discussion using the framing questions I have provided.

Online Module 2	**Topics and Assignments**

Readings (1)	• As background material for our discussion in the next in-person session, and to inform your work on the Photo Diary and Image Board exercises, read:
	  o The article *Margaret Mead Meets Consumer Fieldwork*, Harvard Management Update, August 2001; and,
	  o Chapter 5 - *Identifying Customer Needs* in Ulrich and Eppinger's *Product Design and Development* textbook.
	• You will get the most out of the exercises if you do the background reading first, or at least early in the week as you keep your Photojournal.

Cognitive Design	• View the videos on Affordances and Signifiers before the assigned reading.
	• The videos are not a substitute for the readings in *The Design of Everyday Things* (DOET), but if viewed first, they will aid in the reading.
	• Now read the first two chapters of Don Norman's *The Design of Everyday Things*:
	  o Psychopathology of Everyday Things, Ch. 1
	  The Psychology of Everyday Actions, Ch. 2
	• Be sure that you understand the concepts introduced, and how they relate to each other.
	• Notice that they form natural pairings:
	  o Affordance – Signifier
	  o Mental Model – System Image
	  o Gulf of Execution – Gulf of Evaluation
	• …and that within each pair, the two terms are both related and distinct from each other. Each pair relates to the others as well. How is this?
	• Be prepared to discuss this in our next in-person session, and use what you have learned from the videos and readings to inform your work in the exercises within this module.

Readings (2)	• In *The Design of Everyday Things*, read:
	  o Ch. 3: Knowledge in the Head and in the World; and,
	  o Ch. 4: Knowing What to Do: Constraints, Discoverability, and Feedback.
	• Consider carefully the concepts discussed in these chapters. They represent the next level of understanding the application of Cognitive Design to real-world products, and are vital guides to effective application of the

	foundational concepts of Signifier, Affordance, Mental Model, System Image, Feedback, and the Gulfs of Execution and Evaluation. • Also be sure to prepare the exercise to evaluate a computer application that you commonly use after you have done the reading and are comfortable that you understand the concepts. The exercise will help you solidify your understanding in application.

Design-a-Food Iteration	• Consider whether, and if so how, the concepts introduced in DOET might influence what your team came up with in the Design-a-Food exercise from the previous module.
	  o For example, what affordances does your user want or need?
	  o What signifiers did you create to guide them to notice and use the affordances?
	• Sketch your revised product, submit both the original and the iterated version here.
	• Be prepared to present using a single slide in our next in-person session, explaining your rationale and the revised product; how and why has the product evolved?

Photojournal	• As we discussed in our in-person session, please keep a diary of your activities around either Relaxation or Transportation.
	• Use the templates below, submit on Canvas, and have the materials available in our next in-person session for us to use in an exercise.
	• There are examples and further details in the templates. On Canvas
	• Remember that you need to work on this regularly over the course of an entire week. As you near the end of the week's diary, consider these questions:
	  o Is this “ethnography?” How accurate and actionable do you think the information gained actually is?
	  o How does this relate to traditional marketing research, as you may have learned in your other coursework or practiced in an outside organization?
	  o How does it compare to the approaches Ulrich and Eppinger describe?

Image Board	• Bring in a dozen or so images, from whatever source (online, etc.), that somehow are meaningful to you on the topic (either Relaxation or Transportation) that is not the one you Photojournaled. (So, if you kept a journal on Transportation, bring in images on Relaxation, and vice-versa.) • You do not need to submit, but please have them available in class, as we will construct and analyze image boards in class.
Computer Application Exercise	• View my video reviewing the topics of Cognitive Design and Heuristic Analysis before you attempt this exercise. • As an exercise to test and solidify your understanding of Cognitive Design concepts, think about a computer application you commonly use, then:

	- o Come up with a list of tasks you perform frequently, and that you perform infrequently. - o Run through those tasks, and notice how you know: a) what to do, b) how to do it, and c) whether it has been done as you wanted. - o Take notes on this study exercise, preparing as if you needed to brief a co-worker about your observations. - We will discuss your observations and insights in class, so bring them in a form you can work with (you will not need to present, but will need to refer to them in an exercise). - Does this remind you of anything we have learned previously?
Aibo Case	- Thoroughly prepare, and be ready to discuss in our next in-person session, the case: *Sony AIBO: The World*'s *First Entertainment Robot*, HBS 9-502-010 - As you prepare, consider the following questions: - o Why has AIBO been such a hit? (or has it been?) - o Who are its customers, and why are they buying it? Is there a difference between the US and Japan? - o Does AIBO fit with the rest of Sony's products? - o What drove Sony's definition of what AIBO is and does, what AIBO is not and does not do? - o Would you have defined AIBO the same way? If not, how would you have defined it? - o Where the heck are they going with this?
The Black Ships Film	View my video presenting the Eames film *The Black Ships* and discussing how the imagery that people choose (consciously and subconsciously) can help us understand their thoughts, feelings, and needs around a particular product.
Parking: Experiential Learning → Service Design	- This is an individual assignment, not a team assignment. Please conduct this exercise by yourself, and record your observations in a form that you can work with (you will not be presenting). - o As you interact with parking, on campus, at work, running errands, or just out and about over the next week, pay particular attention to things that work well and things that do not work well at all. - o To help capture the less common edge cases and prepare yourself to observe, use the Critical Incident Technique that we discussed in our previous in-person session to explore your prior experience with parking. - o Then make a special trip to somewhere you know that is difficult to park - there is no need to interact with any other person for this exercise. - o We will draw upon the understanding that you collect for an in-class exercise with your final project team. - As opposed to last quarter, you will not draw upon that understanding for “big picture” ideation. Instead, I will give you a design brief; we will learn and practice tools for designing services and digital interfaces

In-Person Session 2 End of Week 6	**Topics and Preparation**
	We will provide details and guidance, and lead exercises and discussions live in the classroom. Be sure that you have completed all of Online Module 2.
Albo Case Discussion	We will discuss the case *Sony AIBO: The World's First Entertainment Robot, HBS 9-502-010*, and use it to explore the second of the two fundamental approaches to New Product Development. Be sure that you have prepared for the discussion using the framing questions I have provided.
$10 Bill Exercise	We will conduct a practical (and fun!) exercise to solidify our understanding of *Knowledge in the Head* and *Knowledge in the World*.
Computer Application Exercise	In your final project teams, you will begin to apply the Cognitive Design concepts we have learned to a digital product. You will continue building upon this in Online Module 3.

park.ly app	We will conduct this exercise during our in-person session, these instructions are provided solely to help you prepare.
	• Using the understanding of parking that you have gained from your experiential research, storyboard a customer journey for a great parking experience, and wireframe a parking app to serve as a digital touchpoint for the experience, that covers at least the stages of:
	  o Finding an available parking spot
	  o Paying for that parking spot
	  o Remembering where you parked, and how long you own the spot
	• You may decide that the app needs to do more than this - that is fine, if it makes sense.
	
	• Begin building your simple prototype parking app by:
	  o Storyboarding the entire parking process / experience
	  o Building up the Content Inventory
	  o Constructing an Information Architecture
	  o Sketching simple Wireframes (at least 10)
	• We will share, critique, and refine the customer journey and wireframes in class, and then your final project team will build it out into a phone-based prototype during the next online module.
	• Do not waste your time on non-value-creating activities such as account creation, login, shopping carts, payment, etc.
	Focus on the user's parking-related needs.

	• Draw upon the Cognitive Design principles we have learned to inform your app design: o What knowledge does a person need in order to use your app? Should that knowledge reside in their head, or in the world? o What actions should your app afford? How will you signify them? o How will you help people discover what the app will do?

Online Module 3	**Topics and Assignments**
Readings	• Read Chapter 14 in Ulrich and Eppinger's *Product Design and Development* text. • We discussed much of this in our previous in-person session; this is to remind you of our discussions, and assist you in thinking through your own prototyping exercise for this module.
Digital Prototyping	• View my video providing an overview of the tools and process for prototyping digital interfaces and applications. • Pay particular attention to the importance of low-fidelity prototyping, how it leads into higher-fidelity prototyping, and how organizations can use proper sequencing of the two to reach a final prototype faster and at lower cost than their competitors. • Notice how the Customer Journey Map and Storyboarding tools you have already learned feed seamlessly into the Wireframing and Information Architecture layout that are essential steps in digital prototyping. • Consider how moving to an interactive click-through prototype too early can be a waste of resources and lock a team into a sub-optimal design. • What do you think of the framing of the Sketch-to-Prototype continuum? How might you apply this in your own work?
Prototyping Strategies	• View my video discussing the effective employment of physical and digital prototyping methods. • Pay particular attention to the three essential functions of a prototype, and how these feed into the construction of an efficient prototyping strategy.
Manufacturing Methods	• View my video explaining and illustrating the various manufacturing methods employed to produce physical products in large (and small) volumes. • Consider how a robust understanding of your organization's own manufacturing capabilities might enhance or limit your team's design and development efforts.
park.ly iteration:	• As we discussed in our last in-person session, revise your storyboards and wireframes based upon the critique you received, as well as your own heuristic evaluation of your first iteration. Use the tools of Information

heuristics and prototyping	Architecture and Content Inventory to inform your work. (You will not need to present these, but they will be vital to doing good work, and you do need to submit them on Canvas.)
	• For a more detailed approach to setting up and conducting a Heuristic Evaluation, see the following article (it is short, but full of great, actionable information):
	    o [How to Conduct a Heuristic Evaluation]
	    o [10 Usability Heuristics for User Interface Design]
	• For a fun illustration of the heuristics (as applied to video games), take a look at this article:
	    o [10 Usability Heuristics Applied to Video Games]
	• Thinking back to the first set of articles on heuristics, did you notice that there were deeper explanations of each one that you could read or view, if you wanted to know more? That is a good example of Discoverability.
	• As you iterate on your wireframes, ask yourself whether you have designed in a feature (or features) that will make people **love** your app. If not, what might you add?
	
	• Construct phone-based click-through prototypes as discussed in class using the Prototyping on Paper (PoP) app. You can find it here:
	    o https://marvelapp.com/pop/
	• There are both iOS and Android versions available, and it is free for the scale of projects that we will do in this course.
	• The speed and simplicity of creating interactive demos with this is truly amazing. Please use this for the next iteration of your park.ly app, and if you would like to explore more advanced digital prototyping tools for you final project, I have gotten us academic licenses of tools like Balsamiq that are in standard use.
	• Before you begin the process of transforming your revised wireframes into a click-through prototype, watch the video on Canvas from the Google design team. It puts several different prototyping techniques into context, and is a nice illustration of the transition from wireframes, to various paper prototypes, to digital prototypes.
	• Finally, consider what physical products or infrastructure would be required or beneficial to deliver the experience you intend to provide for users. We will use this as an input to the physical prototyping exercise.
	
	• Deliverables:
	    o Revised Content Inventory (upload here)
	    o Revised Information Architecture (upload here)
	    o Revised Wireframes (upload here)
	    o Click-through prototype on a phone, built in PoP (present in our final in-person session)

Network / Ecosystem Analysis: Product I Love	• Bring to our next in-person session a product you **love** – one that fits your life so well that you cannot imagine being without it: • Map out the network / ecosystem for the product. Upload here, and be prepared to show in class a graphic representation of the network and

	discuss why you love the product. Try to link specific features of the product to its fulfillment of your needs / desires. • We have studied several different models and frameworks for relating a person's happiness with a product to the features it incorporates and how those features are actually implemented – these tools are all complimentary to each other, so think about how you might draw upon them to help explain this Product You Love • There are examples of this from previous classes on Canvas.
Individual Team Meeting Week 8/9	• As a team, meet with the professor for a feedback and advice session, to continue helping you work towards your final project product and presentation.

Final Project Product and Presentation	• View my video discussing the purpose and elements of a New Product Development checkpoint, and expectations for the Final Presentation.
	• As we discussed in the previous in-person session, work towards the design and prototype of a product that has both a physical and digital component.
	• You must construct prototypes of both the physical and digital components, test them with at least one target user, analyze the results of that test, and summarize your plans for a next iteration of the product.
	• As a team, craft a presentation of your final project product.
	    o Include, as appropriate, the insights, positioning, testing, and learnings that support your concept as desirable and viable.
	    o Submit your presentation on Canvas.
	• There are three example videos of Final Project presentations from previous classes on Canvas - all classmates working and presenting in a remote online format.
	    o Interestingly, all decided to pursue the product side of their New Venture Design ideas, and (from my perspective) significantly advanced their thinking and ability to represent the product in a compelling way.
	    o Do have a look, I think they are both enjoyable and informative.
	    o I have included the discussion that followed their presentations

In-Person Session 3 End of Week 9/10	**Topics and Preparation**
	We will provide details and guidance, and lead exercises and discussions live in the classroom. Be sure that you have completed all of Online Module 4.
park.ly iteration	In your teams, you will deliver your final iteration of the park.ly app, a maximum of 5 minutes, live to the instructors and your classmates.

	This is separate from your Final Project.
Final Presentations	In your teams, you will deliver your Final Project Product Presentation, live to the instructors and your classmates. You will have as much time to present as you need; most teams find 10-15 minutes to be sufficient.
Wrap-up	We will wrap-up the quarter, tying together both the specific tools we have learned and the overall framework of New Product Development, and discuss how you can integrate this into your own career plans.

Optional Activity	Topics	Reading Assignments
In conjunction with the in-person MGT 452 section	Field Trip: Ascential	
Transportation	**Carpool** – We will meet at Ascential facilities in 4S Ranch / Rancho Bernardo. Our visit starts promptly at 10am. They are generously treating us to lunch. **You should plan on a 45 minute drive to Ascential from Rady in typical morning commute traffic.** **Our visit starts promptly at 10am.**	
Prep Exercises	**Design and Engineering Process / Prototyping / Manufacturing** We will be visiting https://mls.ascentialtech.com **Take some time beforehand to explore their website** – it is quite informative, about process, capabilities, and portfolio, representative of the sort of partner you would consider working with in a real-world NPD effort. Think upon our discussion of prototyping, your reading on the topic, and the larger design and development process that we have been learning. What do you still want to understand better? What do you want to see in action? What do you want to know more about? **Come to the field trip with an active curiosity, primed by reflection on the questions above, and informed by your background research on the company.**	
